



Google Maps for the universe.

Explore the universe through a familiar, interactive map.

THE PROBLEM

Space has coordinates.
It never had directions.

- 1 Astronomy data is public — and still hard to feel.
- 2 Scale is unintuitive. Polaris is 433 light-years away.
- 3 Everyone already knows how to use a map.



Pick anywhere. Get directions. Go.

SWITCH TO LIVE DEMO NOW

Earth

✕

Polaris

✕

+ Add stop

Travel mode

Light speed

299,792 km/s (c)

▼

STRAIGHT-LINE CRUISE · LIGHT SPEED

433 years

433 light-years · 133 pc

Light speed · 299,792 km/s (c) · positions on 2026-10-04

about 5.4 human lifetimes (80 years each)

Distance uncertainty ± 6.4 ly

Begin journey

Camera travels the route · compressed visualization

▶

0 seconds

10 s ▼

⌵

The preview has its own clock (10 s for the whole trip); Play time is paused while it runs.

► Sources and model details

Suggested stops · added distance

Vega

+9.94 light-years (+2.3%)

On the way

+ Add

Capella

+12.8 light-years (+3.0%)

On the way

+ Add

Van Maanen's Star

+12.8 light-years (+3.0%)

On the way

+ Add

01 SEARCH

Any supported destination

02 DIRECTIONS

Earth → Polaris, 433 years

03 TRAVEL MODES

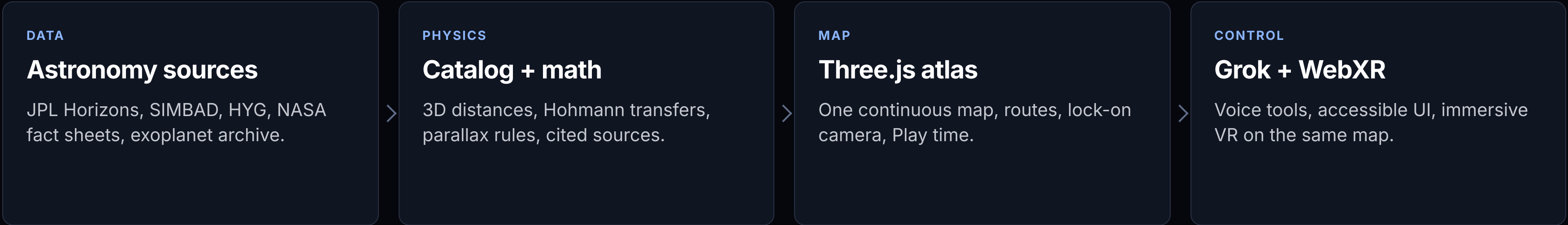
Light speed or Voyager 1

04 MISSION CONTROL

Ask Grok. Tools, not guesses.

HOW IT WORKS

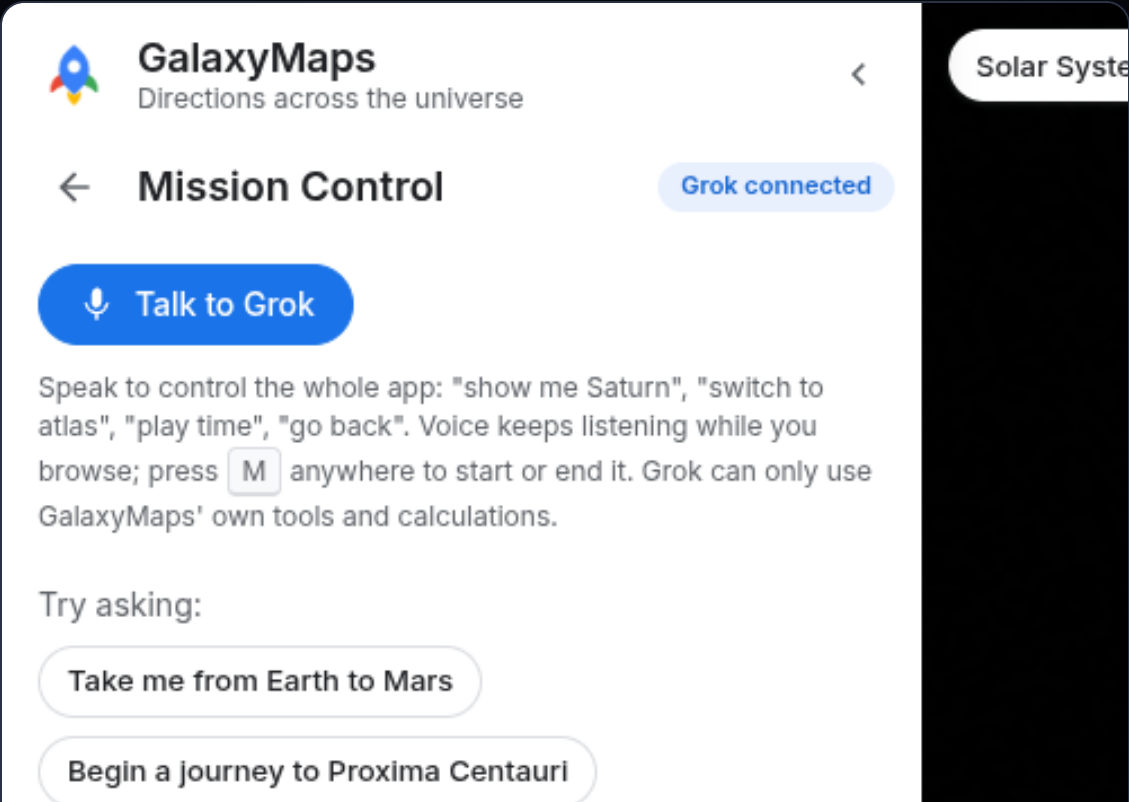
Real space data in. A navigable universe out.



AI explains and controls. Code calculates.

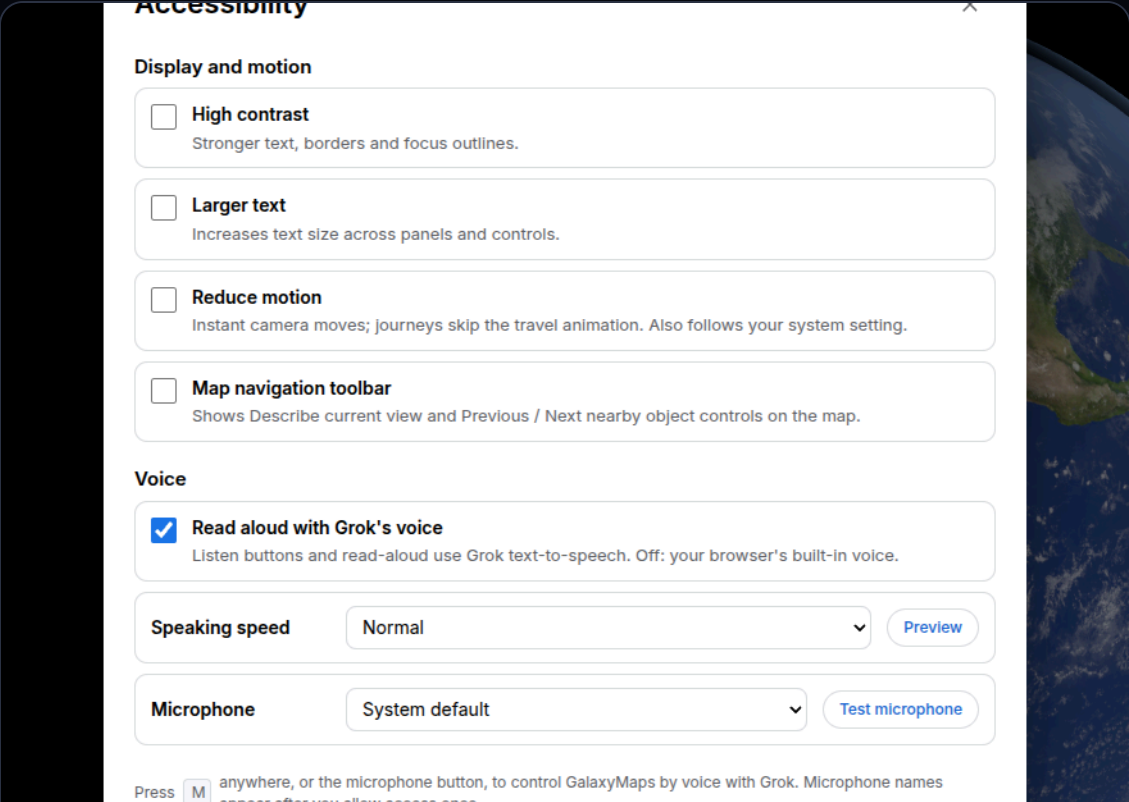
WHY IT IS DIFFERENT

A map you can see, hear, and step inside.



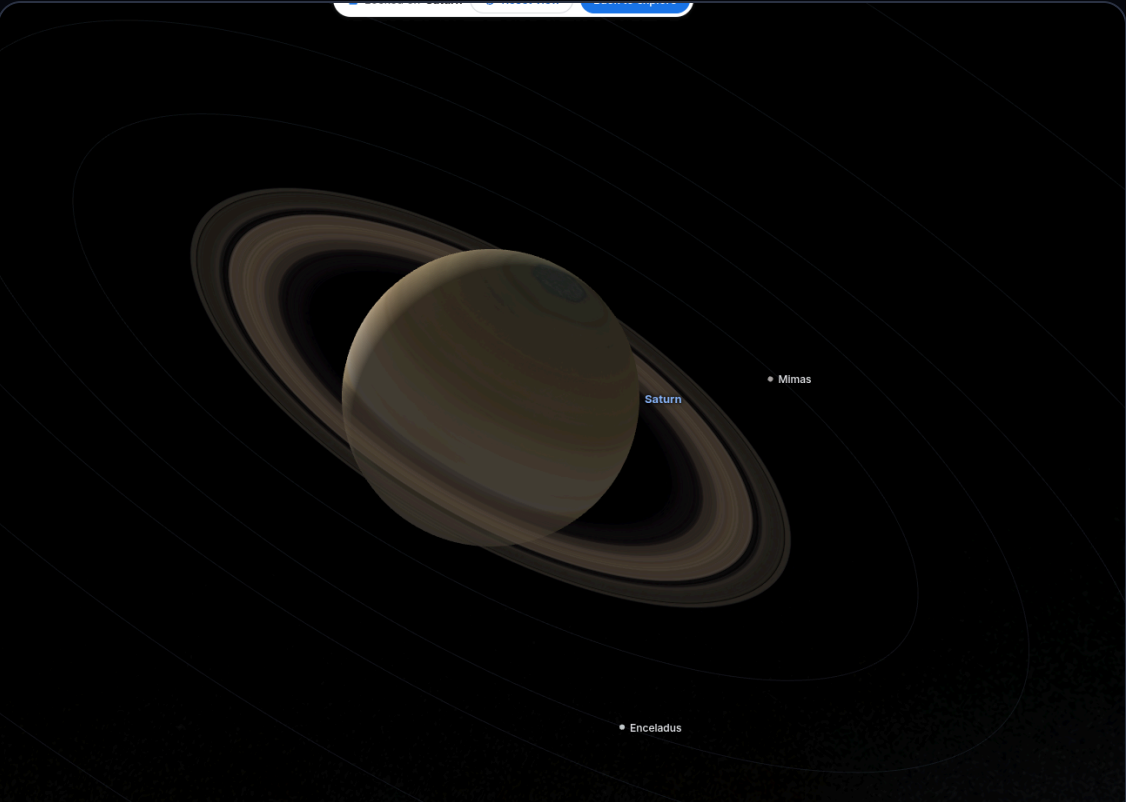
Conversational navigation

Grok Mission Control talks and steers the map through validated tools only.



Accessible exploration

Keyboard, screen reader, voice, and reduced-motion paths — built toward WCAG 2.2 AA.



Immersive navigation

WebXR uses the same 3D map: gaze, pinch, and Grok Voice in VR.

Orion Nebula

GALAXYMAPS

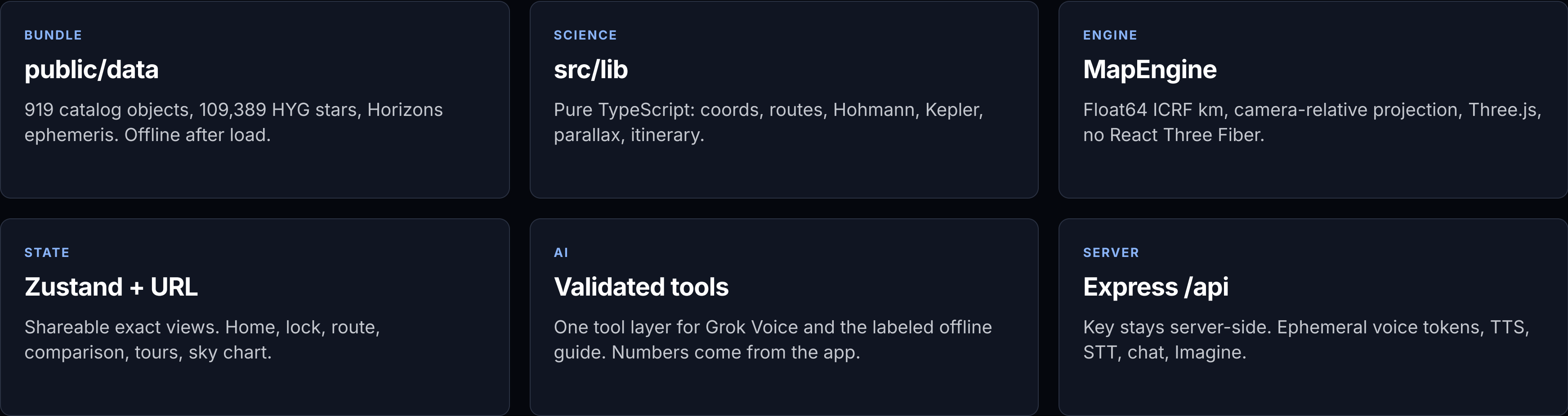
The universe is hard to comprehend. It shouldn't be hard to explore.

GalaxyMaps — Directions across the universe.



Open the repo
github.com/diyar-niyazov/GalaxyMaps

Architecture, from the repository



Data and scientific integrity

Where numbers come from

- JPL Horizons DE441 — Solar System vectors
- SIMBAD — featured stars and deep sky
- HYG v4.4 — 109,389 star positions
- NASA fact sheets + exoplanet archive
- Cited literature for key distances

What we will not fake

- Parallax inverted only if $\sigma/\varpi \leq 20\%$
- No routes to cosmological-redshift objects
- Images labeled observed / illustration / AI
- Milky Way disk is a reconstruction
- Hohmann is idealized, not a launch plan

Earth → Mars

Idealized Hohmann transfer: about 259 days (8.5 months), with Δv , next window, and a separate straight-line benchmark.

Earth → Polaris

True 3D separation ≈ 433 ly. 433 years at light speed; about 7.7 million years at Voyager 1’s measured speed.

Accessible paths and WebXR

Built toward WCAG 2.2 AA

- Keyboard map: / search, H home, D directions, M voice
- Skip links, live regions, ARIA combobox search
- High contrast, larger text, reduced motion
- Describe current view + nearby-object stepping
- We do not claim ADA certification

WebXR on the same map

- Enter VR from a user gesture when supported
- Nearby linear scale; far objects log-compressed
- Head gaze, pinch to fly, dwell to open details
- Grok Voice starts with the session
- Physical Vision Pro / Quest pass still pending

Grok controls the map. It does not invent the sky.

What Grok actually does

- Speech-to-speech over xAI realtime WebSocket
- 5-minute client secret; API key never in the browser
- Can search, route, tour, zoom, play time, share
- Must quote tool-formatted distances and times
- If the API is down: labeled offline scripted guide

SpaceX challenge, honestly

- Built with Cursor
- Uses Grok Voice (and Imagine, labeled AI reconstruction)
- Uses real space / mission catalog data
- No endorsement by SpaceX, xAI, NASA, or Google
- This laptop: Mission Control shows Grok connected